COMPS-B | PROJECT PRESENTATION

Multimodal Accessibility Tool for Visually Impaired People

An AI-powered, voice-driven screen reader and navigation assistant

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PROBLEM STATEMENT

The Digital Barrier

Scale of the Problem

285 million+ people worldwide are visually impaired *(WHO, 2023)*

Limitations of Existing Tools

Existing screen readers only read plain text – they cannot understand images, buttons, or layouts

Independence Gap

Users cannot independently fill forms, click buttons, or navigate modern websites

Cost Barrier

Licensed tools like JAWS cost ₹50,000+ – unaffordable for most users

Missing Integration

No free tool combines visual understanding, speech, and automated UI interaction

OBJECTIVES

What We Set Out to Achieve

01

Capture and understand the full visual context of any computer screen using AI

02

Narrate the screen in natural, human-like language to the user

03

Accept a voice command after narration and interpret user intent

04

Locate the requested UI element on screen using OCR

05

Automatically perform the action (click) on behalf of the user

06

Build a free, lightweight tool that works on any Windows PC

METHODOLOGY

System Design & Approach

Step 1 — Screen Capture

- Triggered by hotkey (Ctrl + Alt + S)
- Screenshot captured and compressed

Step 2 — AI Vision Analysis

- Screenshot sent to a Multimodal Vision LLM (Google Gemma 4 via OpenRouter)
- AI prompted to describe screen like a human guide – layout, buttons, fields, context

Step 3 — Text-to-Speech Narration

AI description spoken aloud using Windows native SpeechSynthesizer

Step 4 — Voice Command Input

- Microphone activates automatically after narration ends
- User speaks one command (e.g. *"Click the login button"*)
- Speech captured and transcribed using Google STT

Step 5 — Intent Parsing + OCR Grounding

- AI extracts target element name from the voice command
- Windows Native OCR scans the screenshot and finds the element's screen coordinates

Step 6 — UI Automation

- Mouse moves to the coordinates and clicks the element (PyAutoGUI)
- System speaks: *"Okay... Done"*

IMPLEMENTATION

Technologies Used

Module	Technology
Screen Capture	Python – Pillow (PIL)
Multimodal AI	Google Gemma 4 (OpenRouter API – Free)
Text-to-Speech	Windows PowerShell SpeechSynthesizer
Speech-to-Text	Google STT + SoundDevice
OCR (UI Grounding)	Windows Native OCR (WinRT)
UI Automation	PyAutoGUI
Hotkey Trigger	Keyboard library
Configuration	Python-dotenv

Key Implementation Highlights

- Custom SoundDeviceMic adapter – bypasses PyAudio Windows build errors
- One-shot listener design – captures one command, acts, then shuts off
- OCR used for spatial grounding instead of asking AI for pixel coordinates (more accurate)
- Entire solution runs on free APIs – zero cost to the user

RESULTS

What the System Achieved

✅ Screen Narration

Screen narrated in natural human language within **~10 seconds** of hotkey press

✅ Voice Command

Voice command transcribed and element located in **under 3 seconds**

✅ UI Action

Mouse automatically moved and clicked the correct UI element

✅ Verbal Confirmation

System confirmed actions verbally ("Okay" and "Done")

✅ Tested Successfully On

Login pages, search bars, navigation tabs, buttons

Accuracy

Narration Relevance

Highly accurate for standard UI layouts

OCR Element Grounding

Accurate for clearly visible text-based elements

Voice Recognition

Accurate in quiet environments

Future Enhancements

Regional language support, mobile app, offline AI mode, and browser extension planned

CONCLUSION

Summary

- **Successfully built a multimodal AI accessibility tool that sees, speaks, listens, and acts**
- **Combines computer vision, NLP, speech recognition, OCR, and UI automation**
- **Enables visually impaired users to navigate any screen using only their voice**
- **Free, lightweight, and runs on any Windows PC with no special hardware**
- **Supports Digital India Mission — empowering 15M+ visually impaired Indians with free, independent digital access**
- **A meaningful step toward making digital technology truly inclusive**

> *"We did not build a workaround. We built equal access."*

REFERENCES

References

- WHO (2023) – *World Report on Vision*
- OpenRouter API – **openrouter.ai**
- Windows WinRT OCR – Microsoft Docs
- PyAutoGUI – **pyautogui.readthedocs.io**
- SpeechRecognition Library – pypi.org/project/SpeechRecognition

Thank You

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